Heart Disease 10-Year Probability Predictor: Machine Learning Prediction with Ensemble Methods

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I. ABSTRACT

Coronary heart disease (CHD) remains a leading cause of death worldwide, with a vast need for accessible and accurate tools for early risk prediction. This paper proposes a machine learning based web application for predicting an individual's 10-year CHD risk using 15 clinical/demographic features derived from the Framingham Heart Study dataset. This dataset contains records of heart patients from Framingham, Massachusetts. Four models have been trained and evaluated: Gradient Boosting, Random Forest, Neural Network, Support Vector Machine. When addressing class imbalance that is present in the dataset, we applied weighted loss functions across the applicable models. Along with this, an ensemble method that combined all four models through probability averaging was implemented to reduce prediction variance. The application was built with Streamlit, a python library used to convert code into a web-based user interface. This application accepts user inputs across demographic, lifestyle, and clinical test results. It then outputs a risk probability along with a one of four tiers of risk classification (low, moderate, high, very high). The results have demonstrated that the ensemble approach outputs more stable metrics than any individual model. This tool is intended for educational purposes and to highlight the potential of combining machine learning techniques along with clinical datasets to help support healthcare decision-making.

II. INTRODUCTION

Heart disease, the leading cause of death for men and women of most racial ethnic groups, is widespread, costly, but somewhat preventable. Despite this, the CDC has reported that one person dies every 34 seconds from cardiovascular disease, and recent years have seen death counts of 919,032 (Heart Disease Facts 2024). Heart disease is a broad term, but can be boiled down to several categories including coronary artery disease, vascular disease, heart rhythm disorders, congenital heart disease, heart failure, and others. One of the reasons for the vast death counts are the many different causes of heart disease in many different categories. For example, factors can come from medical conditions like high blood pressure, high cholesterol, diabetes, as well as lifestyle risk factors like unhealthy diets, not enough exercise, weight issues, smoking, drinking, and finally factors that are out of a patient's control like sex, age, and family history. With all of these factors, it can feel overwhelming to track. These risk factors combine to make heart disease a widespread and persistent burden across the United States as shown by the CDC's prevalence data below.

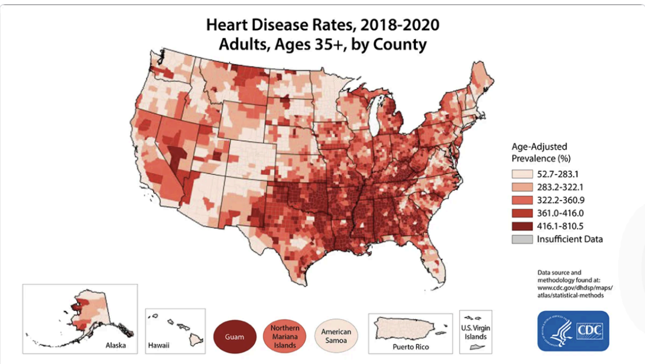

Figure 1: United States map categorizing heart disease rates by county.

While some factors are out of a patient's control, we have researched 15 clinical features to investigate and inform people at risk of coronary artery disease exactly their risk of developing CHD in the next ten years. We have created a dashboard to help the average person identify and track their top risk factors. We also have created a tiered system for levels of risk including low risk (<10%), moderate risk (10 - 19.9%), high risk (20 - 34.9%), and very high risk (>35%). These risk factors are not meant to be a direct diagnosis, but rather inform the patient and encourage proactive conversations with their healthcare provider about prevention and next steps. Ultimately, our goal is to help make heart disease more understandable and actionable for everyday people, and inform personal health thoughts and decisions.

III. LITERATURE REVIEW

A. *Prediction of Heart Disease Based on Machine Learning Using Jellyfish Optimization Algorithm*

The first article reviewed, written by Ahmad Ayid Ahmad and Huseyin Polat, examines the application of machine learning techniques combined with an optimization algorithm to improve the prediction of heart disease. The researchers utilized the Cleveland Heart Disease dataset, which consists of approximately 300 patient records and 13 key features. Although the dataset is relatively limited in size, it provides meaningful real-world clinical data. To enhance model performance, the authors implemented the Jellyfish Optimization Algorithm, a nature-inspired method designed to efficiently identify the most relevant features. This approach reduces unnecessary data complexity while improving the accuracy and reliability of the predictive models.

Following feature selection, the study evaluated multiple machine learning models, including Artificial Neural Networks (ANN), Decision Trees, AdaBoost, and Support Vector Machines (SVM). The results indicated that the SVM model produced the highest level of accuracy, reaching approximately 98.47%, along with strong overall performance metrics. Furthermore, the authors compared their findings with similar studies conducted on different datasets and determined that their proposed method outperformed existing models by approximately 8–20%. Overall, this study highlights the effectiveness of integrating optimization algorithms with machine learning techniques to significantly improve the accuracy and efficiency of heart disease prediction systems.

B. *Heart Disease Prediction Using Machine Learning*

The second article, written by Boukhatem, Youssef, and Nassif, investigates the use of multiple machine learning algorithms to predict heart disease using patient health data. The study utilizes a dataset obtained from Kaggle consisting of 303 instances and 13 attributes, representing common clinical indicators such as age, cholesterol levels, and heart rate. Prior to model development, the data underwent extensive preprocessing, including the removal of extreme outliers, transformation of categorical variables into usable formats, and analysis of feature correlations. Additionally, the dataset was evaluated for class imbalance and found to be relatively balanced, eliminating the need for resampling. Dimensionality reduction techniques, including feature selection through the CfsSubsetEval method and feature extraction using Principal Component Analysis (PCA), were applied to improve model efficiency and reduce redundancy. The dataset was then split into training and testing sets using an 80/20 hold-out method.

The researchers implemented four classification algorithms: Naïve Bayes (NB), Random Forest (RF), Neural Networks (MLP), and Support Vector Machines (SVM). Each model was evaluated using standard performance metrics, including accuracy, precision, recall, and F1-score. Initial results showed moderate performance, but significant improvements were observed after data cleaning and feature selection. Among the models, the Support Vector Machine consistently achieved the highest performance, ultimately reaching an accuracy of 91.67%, along with strong precision and recall values. The study demonstrates that proper data preprocessing and feature selection play a critical role in enhancing model performance. Overall, the findings highlight the effectiveness of machine learning techniques, particularly SVM, in accurately predicting heart disease and emphasize the importance of data quality in developing reliable predictive systems.

C. *A novel approach for the effective prediction of cardiovascular disease using applied artificial intelligence techniques*

The third article, written by Mir et al., presents a comprehensive machine learning framework for predicting cardiovascular disease using multiple datasets and advanced artificial intelligence techniques. Unlike many studies that rely on a single dataset, this research incorporates five distinct datasets, including Heart UCI, Stroke, Heart Statlog, Framingham, and Coronary Heart Disease datasets. The authors designed a three-step methodology consisting of data preprocessing, feature selection, and classification. During preprocessing, missing values were addressed using mean replacement, and class imbalance was handled through sample bootstrapping to reduce bias in the model. Feature selection techniques were then applied to identify the most relevant predictors, improving model efficiency and reducing redundancy.

To evaluate the effectiveness of their approach, the researchers tested multiple machine learning algorithms and ensemble methods, ultimately identifying Random Forest combined with AdaBoost and 10-fold cross-validation as the best-performing model. The results demonstrated exceptionally high accuracy across several datasets, including 99.48% on the Heart Statlog dataset, 93.90% on Heart UCI, 96.25% on the Stroke dataset, 86% on the Framingham dataset, and 78.36% on the Coronary Heart Disease dataset. However, the study also highlights potential limitations, such as the impact of missing data and dataset imbalance on model evaluation, which can lead to overly optimistic accuracy if not properly addressed. Overall, this research emphasizes the importance of robust preprocessing, ensemble learning techniques, and multi-dataset validation in developing reliable and highly accurate cardiovascular disease prediction models.

V. IMPLEMENTATION

A. Data Preprocessing

To begin our implementation of our project, the first natural step was to evaluate potential data sources that might be of use. We first attempted to use a Kaggle dataset, however upon data processing we encountered numerous missing values. The format was so inconsistent that we could not solve the problem by simply dropping rows or columns. In another Kaggle dataset, we ran into another recurring issue in that the data was synthetic in nature. This was not ideal for medical use, as the intention of the project lies in the real world applicability. This leads us to our choice of the renowned Framing Data Set, coming from "The Framingham Heart Study — a long-running cardiovascular cohort study." We performed preprocessing of the data using a target Python script which was mirrored for each training script. The first step of preprocessing after the first inspection was to deal with missing values. We chose to implement two different forms of missing value implementation depending on the data type. For example, there were a total of seven different features that required such action, but we split them into two distinct groups. As for the first group, this contained the amount of cigarettes smoked per day, total cholesterol, BMI, resting heart rate, and fasting blood glucose. Naturally as they are all numerical in nature the method of missing value implementation was to use the median. Furthermore, we utilized similar methods for two categorical features in the highest education level and if the patient is currently taking blood pressure medication. An important note to consider is that Imputers are fit on the training set only, then applied to the test set to prevent data leakage. The next step of action was to properly separate our data into both training and testing sets to accurately evaluate the process and models. We choose to go with an eighty-twenty, training-test split, which is industry standard. Additionally, we chose to perform feature scaling on the appropriate models to best evaluate their performances. The tree based models such as Random Forest and Gradient Boosting did not require scaling, however the SVM and Neural Network approach do require feature scaling.it. Both make use of the sklearn standard scaler package, utilizing zero mean, unit variance fit on training data only. The fitted scaler objects are saved alongside each model so they can be re-applied identically at inference time. We had a total of fifteen different features including *male, age, education, currentSmoker, cigsPerDay, BPMeds, prevalentStroke, prevalentHyp, diabetes, totChol, sysBP, diaBP, BMI, heartRate, glucose,* To help visualize the proper feature importance we present the feature correlation matrix below(Figure 1).

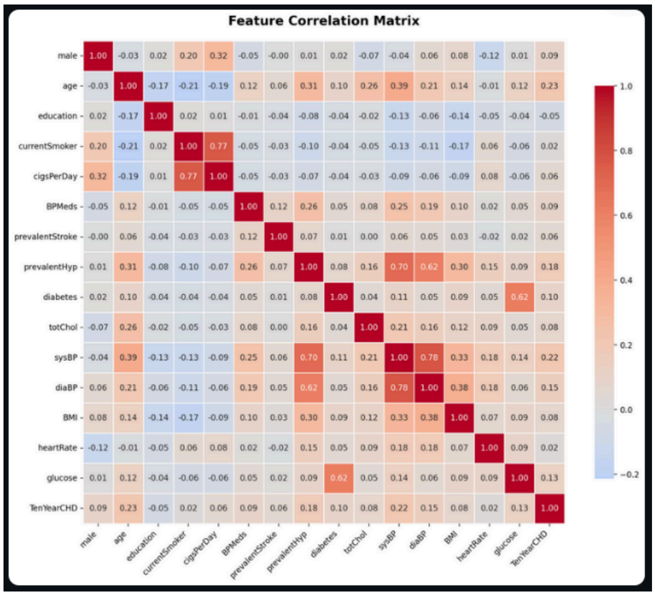

Figure 2: Feature Correlation Matrix from EDA notebook.

B. Model Training

a. Gradient Boosting

The gradient boosting model builds trees sequentially, with each new tree correcting the residual errors of the ensemble until that point. There was no scaler needed for this model since tree splits are invariant to monotonic feature transformations. The parameters associated with training the model were as follows: n_estimators = 200 (200 sequential decision trees), learning_rate = 0.05 (small step size used to reduce overfitting), max_depth = 4 (shallow trees for regularization), and subsample = 0.8 (80% of training data per tree). The performance metrics of this model consisted of:

ROC-AUC = 0.6425
Accuracy = 0.83
Precision = 0.2857
Recall = 0.0775

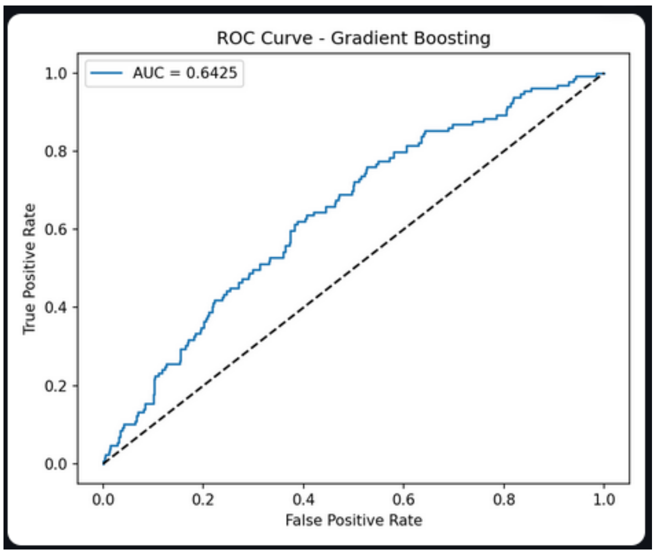

Figure 3: ROC Curve from Gradient Boosting training

b. Random Forest

When performing model training we utilized a python script containing RandomForestClassifier in which there are seven different features. These are *n_estimators, max_depth, min_samples_split, min_samples_leaf, class_weight, random_state,* and, *n_jobs*. Our Random Forest builds 200 trees in parallel, each on a bootstrap sample of the data and a random subset of features. Predictions are averaged over all trees. automatically adjusts sample weights inversely proportional to class frequencies, compensating for the dataset imbalance. The performance metrics of Random Forest model consisted of:

ROC-AUC = 0.6721
Accuracy = 0.7335
Precision = 0.2587
Recall = 0.4031

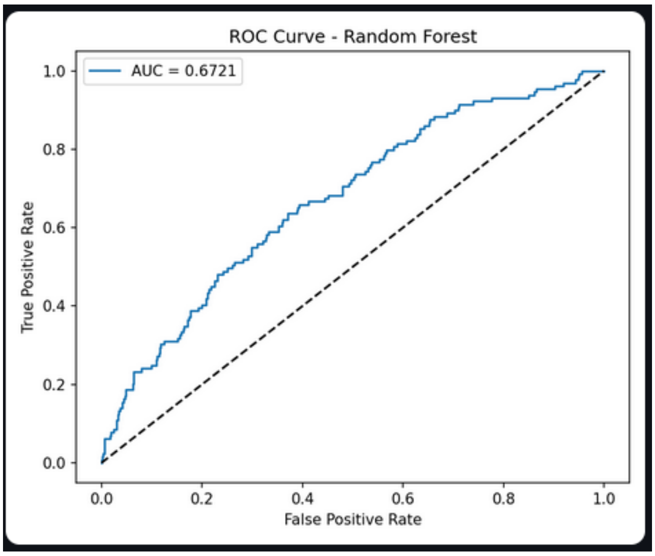

Figure 4: ROC Curve from Random Forest training

c. Neural Network

The architecture of this model takes the 15 input features and passes them through two hidden layers. The first layer projects from 15 to 64 neurons which is then followed by a ReLU activation, batch normalization, and a dropout rate of 0.3. Then the second hidden layer reduces the dimensionality from 64 to 32 neurons and applies that same pattern as the first layer but with a slightly lower dropout rate of 0.2. The final layer maps these 32 neurons into a single output. The neural network was trained for 60 epochs with a batch size of 64 and the Adam optimizer with a learning rate of 1e-3 and weight decay of 1e-4. We utilized a StepLR scheduler to halve the learning rate every 20 epochs, which allowed the model to make finer adjustments during its training process. The class imbalance was handled with the loss function BCEWithLogitsLoss. The performance metrics of the neural network consisted of:

ROC-AUC = 0.6721
Accuracy = 0.7335
Precision = 0.2587
Recall = 0.4031

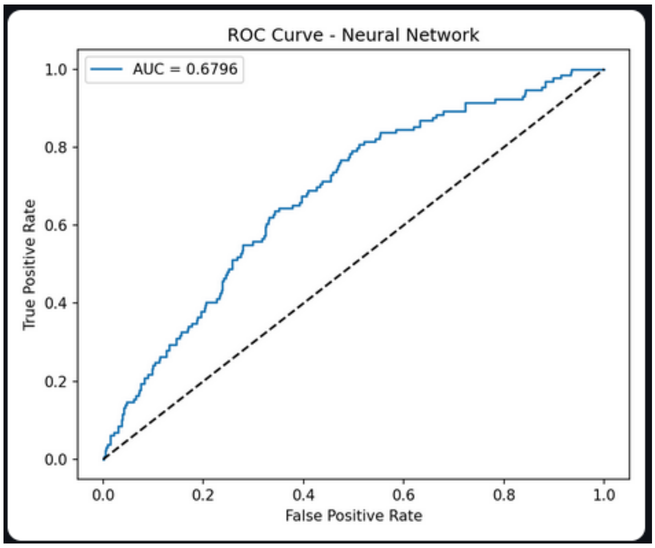

Figure 5: ROC Curve from Neural Network training

d. Support Vector Machine

When performing model training we utilized a python script containing SupportVectorClassification (SVC) as part of SVM in which there are five different features. These are *C, kernel, probability, class_weight,* and *random_state*. SVM finds the maximum-margin hyperplane separating CHD-positive and CHD- negative cases in feature space. The RBF kernel allows it to learn non-linear decision boundaries. Because SVM is distance-based, features must be scaled — the aforementioned StandardScaler is saved and applied at inference time. The probability feature uses Platt scaling (a logistic regression fit on cross-validated scores) to convert the SVM distance-from-hyperplane into a calibrated probability. Feature importance for SVM is computed via permutation importance: features are shuffled one at a time and the mean decrease in ROC-AUC is measured. The performance metrics of the Support Vector Machine consisted of:

ROC-AUC = 0.6705
Accuracy = 0.6792
Precision = 0.2473
Recall = 0.5426

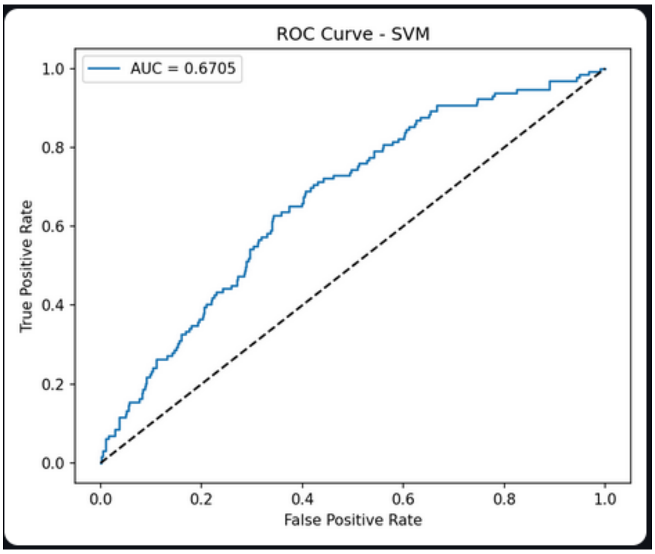

Figure 6: ROC Curve from SVM training

C. Prediction Dashboard

The Prediction Dashboard is the cumulative product of all of our models and features that are available to prospective users. This was created through the use of Streamlit, an open source python framework. The main evaluation metric used is ROC-AUC, which is the scalar value based upon the ROC graph. There are four separate input sections for users to put their information. The different sections are *About You, Lifestyle, Medical History,* and *Vitals & Lab Results*. The dashboard utilizes an ensemble model which is comprised of the four different models. Users choose from: Ensemble (All Models Combined) — recommended, averages all loaded Models Gradient Boosting — individual model Random Forest — individual model SVM — individual model Neural Network — individual model (only available if PyTorch is installed and the .pt file exists). The way the ensemble model works, is that The function ensemble_proba loops over all loaded models, calls predicted probability for each, and returns the arithmetic mean of all probabilities: probs = [predict_proba(name, features) for name in models] return float(np.mean(probs)) This simple averaging reduces variance and generally produces more robust estimates than any single model alone, especially when the individual models disagree. The Risk Tier Classification helps users to intuitively understand and visualize their results, and individualized risk. If they are less than ten percent, a bright green visualization will appear, placing them into the Low Risk Tier. If they are between ten and twenty percent, a yellow visualization will appear, placing them into the Moderate Risk Tier. If they are between twenty and thirty-five percent, an Orange/Red visualization will appear, placing them into the High Risk Tier. If they are greater than thirty-five percent, a bright green visualization will appear, placing them into the Very High Risk Tier.

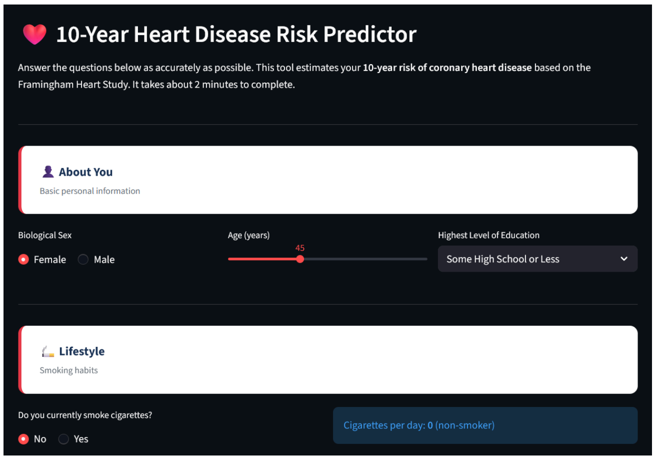

Figure 7: Risk Predictor Application interface

VI: RISK DASHBOARD CASE STUDIES

A. Case 1

The first test case consists of a very healthy female of the age 32. She attended some college, but never graduated and does not smoke cigarettes. Here are her diagnoses and lab results:

BP Medication: No
Prevalent Stroke: No
Prevalent Hypertension: No
Diabetes: No
Total Cholesterol:185mg/dL
Systolic BP: 112mmHg
Diastolic BP: 74mmHg
BMI: 22.4kg/m^2
Heart Rate: 68bpm
Fasting Glucose: 88mg/dL

Her results appeared just as you'd expect, with a 4.3% estimated 10-year CHD probability. This would dictate a low risk and prompt her to continue her healthy habits.

Figure 8: Probability Predictor output from Case 1

B. Case 2

The second case consists of a middle-aged man of the age 42. He is a high school graduate and a current smoker that smokes about 6 cigarettes a day. Here are his diagnoses and lab results:

BP Medication: Yes
Prevalent Stroke: No
Prevalent Hypertension: No
Diabetes: No
Total Cholesterol: 210mg/dL
Systolic BP: 135mmHg
Diastolic BP: 88mmHg
BMI: 25.8kg/m^2
Heart Rate: 78bpm
Fasting Glucose: 98mg/dL

His results were classified in the moderate risk tier with a 19.4% estimated 10-year CHD probability. This prompted the user to talk to his doctor about steps he can take to lower this risk.

Figure 9: Probability Predictor output from Case 2

C. Case 3

The third and final case consists of an older man of the age 62. He did not graduate high school and is a heavy smoker that smokes about 30 cigarettes a day. Here are his diagnoses and lab results:

BP Medication: Yes
Prevalent Stroke: Yes
Prevalent Hypertension: Yes
Diabetes: No
Total Cholesterol: 275mg/dL
Systolic BP: 168mmHg
Diastolic BP: 98mmHg
BMI: 33.2kg/m^2
Heart Rate: 92bpm
Fasting Glucose: 148mg/dL

His results showed he was at a very high risk of 70.5% estimated 10-year CHD probability. The user was prompted to consult a doctor as soon as possible to discuss immediate preventive care.

Figure 10: Probability predictor output from Case 3

VII: LIMITATIONS

While the model demonstrated strong predictive performance, several limitations should be considered. To start, our model is trained on the Framingham dataset which consists of a dominantly white population from Massachusetts. This is not representative of the full population, and may not accurately reflect broader and more diverse groups. Whether it be race, geography, or socioeconomic status, these differences between groups can change the risk patterns and could possibly cause our dashboard to overestimate or underestimate risk. This certainly raises concern for the generalizability of predictions across diverse populations that would be interacting with our dashboard if it were to be implemented in the real world.

The second limitation to consider is that the Framingham dataset is outdated. The data originates from the early 1900s and a lot has changed since then. There are improved medications available, people have different lifestyles and diets, and healthcare is not the same as it was 100 years ago. This is all important to consider because the risk relationships may not hold the same today and our predictions may not reflect modern risk accurately. This limitation may make it challenging to apply in an actual current clinical setting.

The third limitation to think about is the limited feature set. The model only uses the features presented in the dataset, and this may limit the ability to effectively diagnose in a real clinical setting. For example our dataset does not take into account a patient's lifestyle or family history. Things like the kind of food a patient eats, the amount they exercise, or their family history/genetics could be critical in predicting heart disease. Therefore, our predictions are incomplete by nature, and this would limit the model's ability to fully represent heart disease risk.

The last limitation we will address is that our dashboard is not a clinical decision tool. For many different reasons this is not a tool that doctors should directly rely on. It doesn't replace medical judgement and it is important to note that it doesn't take into consideration the context or patient history. This is a very practical limitation and is meant to make it known that our dashboard is not meant to create medical conclusions. With that said, it can still be a powerful tool in creating supportive insights for doctors.

VIII: CONCLUSION / FUTURE WORK

In this paper, we developed a machine learning approach to predict 10-year coronary heart disease using data from the Framingham heart study. By testing multiple models including Gradient Boosting, Random Forest, SVM, and a Neural Network, and combining them into an ensemble, we were able to produce more stable predictions than any single model alone. We also created a dashboard to present these predictions in a way that is easy to understand and apply. Based on the test cases, the model was able to distinguish between different risk levels in a way that generally aligns with expected clinical patterns.

Overall, this project highlights how machine learning can be used to support understanding of heart disease risk. While there are limitations, the results show that combining models and presenting outputs clearly can make these types of tools more useful and accessible.

Future work should focus on expanding beyond the Framingham dataset to include larger and more diverse datasets in order to improve the generalizability of heart disease prediction models. As we talked about in the limitation section, the Framingham dataset represents a specific population and may not fully capture variations across different demographics or healthcare environments. Incorporating additional datasets from hospitals, national health systems, or international studies would allow models to learn from a broader range of patient characteristics. This would help reduce overfitting and improve the reliability of predictions when applied to real-world populations.

In addition to expanding the dataset, future research should further explore and compare different machine learning models. In this study, Gradient Boosting, SVM, Neural Networks, and Random Forest were used to predict heart disease. While these models demonstrated strong performance, testing additional approaches such as more advanced deep learning architectures could lead to further improvements in accuracy.

Finally, incorporating additional features into the dataset could significantly enhance model performance. Although the Firmingham dataset includes key clinical variables, heart disease is influenced by a wide range of factors beyond those currently included. Future studies could expand the feature set to include lifestyle behaviors such as diet and physical activity as well as genetic information and data collected from wearable health devices. By integrating a more comprehensive set of features, models may be able to capture more complex patterns and provide more accurate and personalized predictions, ultimately improving early detection and prevention of CHD.

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